

HIV Testing - Medical Clinical Policy Bulletins

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Policy

Scope of Policy

This Clinical Policy Bulletin addresses HIV testing.

1. Medical Necessity

Aetna considers human immunodeficiency virus (HIV) testing medically necessary for the following:

1. For screening persons for HIV infection, according to the recommendations of the U.S. Preventive Services Task Force and the Centers for Disease Control and Prevention;
2. Initial testing for HIV with a DA-approved antigen/antibody combination immunoassay that detects HIV-1 and HIV-2 antibodies and HIV-1 p24 antigen to screen for established infection with HIV-1 or HIV-2 and for acute HIV-1 infection. No further testing is required for specimens that are nonreactive on the initial immunoassay.
3. Further testing of specimens with a reactive antigen/antibody combination immunoassay result (or repeatedly reactive, if repeat testing is recommended by the manufacturer or required by regulatory authorities), if tested with an FDA-approved antibody immunoassay that differentiates HIV-1 antibodies from HIV-2 antibodies. Reactive results on the initial antigen/antibody combination immunoassay and the HIV-1/HIV-2 antibody differentiation immunoassay should be interpreted as positive for HIV-1 antibodies, HIV-2 antibodies, or HIV antibodies, undifferentiated.
4. Testing of specimens that are reactive on the initial antigen/antibody combination immunoassay and nonreactive or indeterminate on the HIV-1/HIV-2 antibody differentiation immunoassay when tested with an FDA-approved HIV-1 nucleic acid test (NAT) according to the following criteria:
 1. A reactive HIV-1 NAT result and nonreactive HIV-1/HIV-2 antibody differentiation immunoassay result indicates laboratory evidence for acute HIV-1 infection;
 2. A reactive HIV-1 NAT result and indeterminate HIV-1/HIV-2 antibody differentiation immunoassay result indicates the presence of HIV-1 infection confirmed by HIV-1 NAT;
 3. A negative HIV-1 NAT result and nonreactive or indeterminate HIV-1/HIV-2 antibody differentiation immunoassay result indicates a false-positive result on the initial immunoassay.

Note: Laboratories should use this same testing algorithm, beginning with an antigen/

antibody combination immunoassay, with serum or plasma specimens submitted for testing after a reactive (preliminary positive) result from any rapid HIV test.

5. The Orasure oral HIV test kit (OraSure Technologies, Bethlehem, PA) for the same indications as standard HIV testing;
6. The OraQuick Rapid HIV-1 Antibody point-of-care test kit (OraSure Technologies, Bethlehem, PA) as an adequate alternative to laboratory HIV blood tests for medically necessary indications for HIV testing.

2. Policy Limitations and Exclusions

Note: Aetna does not cover home HIV test kits that do not require a physician's prescription under any plans. These include:

- Confide Home HIV Test (Johnson & Johnson) [*](#)
- Home Access HIV Test System.

* The Confide Home HIV Test kit was withdrawn from the market in 1997 due to lack of consumer interest.

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
<i>Information in the [brackets] below has been added for clarification purposes. Codes requiring a 7th character are represented by "+":</i>	
CPT codes covered if selection criteria are met:	
86689	HTLV or HIV antibody, confirmatory test (e.g., Western Blot)
86701	Qualitative or semiquantitative immunoassay performed by multiple step methods for the detection of antibodies to infectious agents; HIV-1
86702	Qualitative or semiquantitative immunoassay performed by multiple step methods for the detection of antibodies to infectious agents; HIV-2
86703	Antibody; HIV-1 and HIV-2, single result
87389	Infectious agent antigen detection by immunoassay technique, (eg, enzyme immunoassay [EIA], enzyme-linked immunosorbent assay [ELISA], immunochemiluminometric assay [IMCA]) qualitative or semiquantitative, multiple-step method; HIV-1 antigen(s), with HIV-1 and HIV-2 antibodies, single result
87390	HIV-1
87391	HIV-2
87534	Infectious agent detection by nucleic acid (DNA or RNA); HIV-1, direct probe technique
87535	HIV-1, amplified probe technique, includes reverse transcription when performed
87536	HIV-1, quantification, includes reverse transcription when performed
87806	Infectious agent antigen detection by immunoassay with direct optical observation; HIV-1 antigen(s), with HIV-1 and HIV-2 antibodies

Code	Code Description
HCPCS codes covered if selection criteria are met:	
G0432	Infectious agent antigen detection by enzyme immunoassay (EIA) technique, qualitative or semi-quantitative, multiple-step method, HIV-1 or HIV-2, screening
G0433	Infectious agent antigen detection by enzyme-linked immunosorbent assay (ELISA) technique, antibody, HIV-1 or HIV-2, screening
G0435	Infectious agent antigen detection by rapid antibody test of oral mucosa transudate, HIV-1 or HIV-2, screening
G0475	HIV antigen/antibody, combination assay, screening
S3645	HIV-1 antibody testing of oral mucosal transudate
ICD-10 codes covered if selection criteria are met:	
B20	Human immunodeficiency virus [HIV] disease
Z11.4	Encounter for screening for human immunodeficiency virus [HIV]
Z20.6	Contact with and (suspected) exposure to human immunodeficiency virus [HIV]
Z20.828	Contact with and (suspected) exposure to other viral communicable diseases
Z21	Asymptomatic human immunodeficiency virus [HIV] infection status

Background

According to the U.S. Preventive Services Task Force (USPSTF, 2005), human immunodeficiency virus (HIV) testing is indicated in individuals with the following risk factors for HIV infection:

- Individuals whose past or present sex partners were HIV-infected, bisexual, or injection drug users; or
- Men and women having unprotected sex with multiple partners; or
- Men and women who exchange sex for money or drugs or have sex partners who do; or
- Men who have had sex with men after 1975; or
- Past or present injection drug users; or
- Persons being treated for sexually transmitted diseases (STDs); or
- Persons with a history of blood transfusion between 1978 and 1985.

Persons who request an HIV test despite reporting no individual risk factors may also be considered at increased risk, since this group is likely to include individuals not willing to disclose high-risk behaviors.

The USPSTF (2005) states that there is good evidence of increased yield from routine HIV screening of persons who report no individual risk factors but are seen in high-risk or high-prevalence clinical settings.

High-risk settings include any of the following:

- Adolescent health clinics with a high prevalence of STDs
- Clinics serving men who have sex with men
- Correctional facilities

- Homeless shelters
- STD clinics
- Tuberculosis clinics.

High-prevalence settings are defined by the Centers for Disease Control and Prevention (CDC) as those known to have a 1 % or greater prevalence of infection among the patient population being served. Where possible, clinicians should consider the prevalence of HIV infection or the risk characteristics of the population they serve in determining an appropriate screening strategy. Data are currently lacking to guide clinical decisions about the optimal frequency of HIV screening.

The USPSTF (2005) recommended that clinicians screen all pregnant women for HIV. The USPSTF made no recommendation for or against routinely screening for HIV in adolescents and adults who are not at increased risk for HIV infection.

In 2006, CDC revised its recommendations for HIV screening of patients in all healthcare settings. Major revisions include a recommendation for routine HIV screening for patients aged 13 to 64 years after the patient is notified that testing will be performed unless the patient declines (opt-out screening). The CDC recommended that persons at high-risk for HIV infection should be screened for HIV at least annually.

The American College of Physicians (ACP) and HIV Medicine Association developed a guidance statement on screening for HIV in health care settings (Qaseem et al, 2009). In accordance to the CDC's 2006 recommendation, the ACP/HIV Medicine Association stated that in all health-care settings, screening for HIV infection should be performed routinely for all patients aged 13 to 64 years. Healthcare providers should initiate screening unless prevalence of undiagnosed HIV infection in their patients has been documented to be less than 0.1 %. In the absence of existing data for HIV prevalence, healthcare providers should initiate voluntary screening until they establish that the diagnostic yield is less than 1 per 1,000 patients screened, at which point such screening is no longer warranted.

Recommendations for HIV screening in pregnant women were revised to include such screening in the routine panel of prenatal screening tests for all pregnant women (CDC, 2006). The CDC recommended that HIV screening after the pregnant woman is notified that testing will be performed unless the patient declines (opt-out screening). The CDC recommended repeat HIV testing in the third trimester for all women at high-risk for HIV, for women in jurisdictions with elevated HIV or AIDS incidence, and for women receiving healthcare in facilities with at least 1 diagnosed HIV case per 1,000 pregnant women per year.

In a Cochrane review, Bateganya et al (2007) discussed home-based HIV voluntary counseling and testing delivery models, which, given the low uptake of facility-based testing models, may be an effective method to get more patients on treatment and prevent new infections in developing countries. The authors concluded that home-based testing and/or delivery of HIV test results at home, rather than in clinics, appears to lead to higher uptake in testing. However, given the limited extant literature and the limitations in the included existing studies, there is insufficient evidence to recommend large-scale implementation of the home-based testing model.

The OraSure oral HIV test kit (OraSure Technologies, Bethlehem, PA) is an HIV test kit that uses oral mucosal transudate for testing rather than blood testing. The kit is manufactured by SmithKline Beecham Consumer Healthcare and is provided to physicians for use in their office. This HIV test kit requires physician involvement and is used for the same indications as standard HIV testing.

The OraQuick Rapid HIV-1 Antibody point-of-care test kit (OraSure Technologies, Bethlehem, PA) is a rapid, point-of-care test designed to detect antibodies to HIV-1 in fingerstick whole blood and venipuncture whole blood specimen within approximately 20 mins. In the clinical studies submitted to the U.S. Food and Drug Administration (FDA) by the manufacturer, OraQuick correctly identified 99.6 % of people who were infected with HIV-1 (sensitivity) and 100 % of people who were not infected with HIV-1 (specificity). As with all HIV screening tests, a reactive test result needs to be confirmed by an additional, more specific Western blot test.

In a Cochrane review, Bateganya et al (2010) examined the effect of home-based HIV voluntary counselling and testing (VCT) on uptake of HIV testing. These investigators

searched MEDLINE (February 2007), EMBASE (February 2007), CENTRAL (February 2007), AIDSearch (February 2007), LILACS, CINAHL and Sociofile. They also contacted relevant researchers. The original review search strategy was updated in 2008. Randomized controlled trials comparing home-based HIV VCT with other testing models were used for this analysis. Two review authors independently selected studies, assessed methodological quality, and extracted data. They planned to conduct statistical analysis using the Review Manager software and calculate summary statistics (relative risks (RRs) with 95 % confidence intervals (CI)) for primary outcomes. Only 1 study from developing countries met the inclusion criteria and was included in the review. The study, a cluster randomized trial (10 clusters, n = 849) compared VCT uptake between an optional location (including home-based) and a local clinic location in a population-based HIV survey. The study showed a higher uptake of VCT among participants in the optional-location group. Uptake was significantly greater in the optional-location group in those who were pre-test counselled only (RR = 4.6; 95 % CI: 3.58 to 5.91); pre-test counselled and tested (RR = 4.6; 95 % CI: 3.51 to 5.92); and post-test counselled and received the test result (RR = 4.8; 95 % CI: 3.62 to 6.21). This study, however, had significant methodological problems limiting further analysis and interpretation. The authors concluded that although home-based HIV VCT has the potential to enhance VCT uptake in developing countries, insufficient data exist to recommend large-scale implementation of home-based HIV testing. They stated that further studies are needed to determine if home-based VCT is better than facility-based VCT in improving VCT uptake.

The USPSTF released updated recommendations for HIV screening in April, 2013, including two Grade A recommendations. The USPSTF recommends that clinicians screen for HIV infection in adolescents and adults ages 15 to 65 years. Younger adolescents and older adults who are at increased risk should also be screened. The USPSTF also recommends that clinicians screen all pregnant women for HIV, including those who present in labor who are untested and whose HIV status is unknown.

In June, 2014 the CDC updated recommendations for screening for HIV to include an algorithm for a sequence of tests used in combination to improve the accuracy of the laboratory diagnosis of HIV. The recommendations concern testing of serum or plasma specimens. Previous CDC recommendations only employed tests for HIV antibodies, but the updated recommendations also include tests for HIV antigens and HIV nucleic acid. The recommendations have been issued because studies from populations at high risk for HIV demonstrated that antibody testing alone may miss a considerable percentage of HIV infections detectable by virologic tests. Advantages of these updated recommendations include improved accuracy of laboratory diagnosis of acute HIV-1 infection, equally accurate laboratory diagnosis of established HIV-1 infection, improved accuracy of laboratory diagnosis of HIV-2 infection, a decrease in the number of indeterminate results, and faster turnaround time for most test results (CDC, 2014).

Muthukumar et al (2015) evaluated the false-positive and false-negative rates of a fourth-generation HIV assay, the Abbott ARCHITECT, versus 2 HIV third-generation assays, the Siemens Centaur and the Ortho-Clinical Diagnostics Vitros. These researchers examined 123 patient specimens. In the first phase of the study, they compared 99 specimens that had a positive screening result via the third-generation Vitros assay (10 positive, 82 negative, and 7 indeterminate via confirmatory immunofluorescent assay [IFA]/Western blot [WB] testing). In the second phase, they assessed 24 HIV-1 RNA-positive (positive result via the nuclear acid amplification test [NAAT] and negative/indeterminate results via the WB test) specimens harboring acute HIV infection. The fourth-generation ARCHITECT assay yielded fewer false-positive results (n = 2) than the third-generation Centaur (n = 9; p = 0.02) and Vitros (n = 82; p < 0.001) assays. One confirmed positive case had a false-negative result via the Centaur assay. When specimens from the 24 patients with acute HIV-1 infection were tested, the ARCHITECT assay yielded fewer false-negative results (n = 5) than the Centaur (n = 10) (p = 0.13) and the other third-generation tests (n = 16) (p = 0.002). The authors concluded that the findings of this study indicated that the fourth-generation ARCHITECT HIV assay yields fewer false-positive and false-negative results than the third-generation HIV assays we tested.

An UpToDate review on "Screening and diagnostic testing for HIV infection" (Bartlett, 2015) states that "Combination HIV antigen and antibody tests - Fourth generation HIV tests are distinguished from antibody-only tests by their ability to detect both antibody and HIV p24 antigen. Of the available tests in the United States, two are designed to provide rapid results (i.e., within 30 minutes)":

- The ARCHITECT HIV Ag/Ab Combo test is an immunoassay that is designed to be performed in

the laboratory. It can be used to analyze batches of specimens or to provide rapid results on a single sample.

- The Determine HIV 1/2 Ag/Ab Combo provides rapid results using lateral flow technology on a single specimen. In the United States, the Determine HIV 1/2 Ag/Ab combo is only approved for use in the laboratory. However, since lateral flow technology combines all the reagents into a single test strip, this test is also well suited for point-of-care settings. However, a Clinical Laboratory Improvement Amendment (CLIA) waiver to use this test in point-of-care settings is pending.

HIV Home-Based Testing / Self-Testing

Frye and co-workers (2015) noted that HIV disproportionately affects young black men who have sex with men (MSM) and transgender women in the US. Increasing HIV testing rates among these populations is a critical public health goal. Although HIV self-tests are commercially available, there is a need to better understand access to and uptake of HIV self-testing among this population. These investigators reported results of a qualitative study of 30 young black MSM and transgender women residing in the New York City area to understand facilitators of and barriers to a range of HIV testing approaches, including self-testing. Mean age was 23.7 years (SD = 3.4). Over half (54 %) had some college or an associate's degree, yet 37 % had an annual personal income of less than \$10,000 per year. Most (64 %) participants had tested in the past 6 months; venues included community health/free clinics, medical offices, mobile testing units, hospitals, emergency departments, and research sites. Just 1 participant reported ever using a commercially available HIV self-test. Facilitators of self-testing included convenience, control, and privacy, particularly as compared to venue-based testing.

Barriers to self-testing included the cost of the test, anxiety regarding accessing the test, concerns around correct test operation, and lack of support if a test result is positive. Participants indicated that instruction in correct test operation and social support in the event of a positive test result may increase the likelihood that they would use the self-test. The authors concluded that alongside developing new approaches to HIV prevention, developing ways to increase HIV self-testing is a public health priority for young, black MSM, and transgender women.

Estem and associates (2016) stated that oral HIV self-testing is an innovative and potentially high-impact means to increase HIV-case identification globally. As a screening test, oral HIV self-testing offers the potential for increased adoption through greater convenience and privacy, and the potential to increase the proportion of the population who test regularly. Research on how best to translate the innovation of oral self-testing to high-risk populations is underway. Presently, only 1 oral HIV self-test kit is FDA-approved (OraQuick In-Home HIV Test) and available for retail sale.

Kenya and colleagues (2016) noted that 60 % of African Americans have had an HIV test, yet this population disproportionately contributes to AIDS mortality, suggesting that testing is not occurring early enough to achieve optimal outcomes. OraQuick, an FDA-approved home-based HIV rapid test (HBHRT), could potentially increase testing rates.

These investigators examined if community health workers (CHWs) paired with HBHRT could improve HIV screening and health care access among African Americans in Miami, Florida.

In October-November 2013, a total of 60 African Americans were enrolled and randomized to the experimental condition, which received CHW assistance to complete HBHRT, or the control condition, which were instructed to complete HBHRT independently. Intervention participants were significantly ($p \leq 0.05$) more likely than control participants to complete HBHRT and, if positive, get linked to HIV care (100 % versus 83 %) $\chi^2 (1, n = 60) = 5.46, p \leq 0.02$. The authors concluded that CHW-assisted HBHRT may be a promising strategy to improve HIV testing and care among African Americans.

The OraQuick Advance Rapid HIV-1/2 Test

Curlin and colleagues (2017) noted that the OraQuick Advance Rapid HIV-1/2 Test is a point-of-care test capable of detecting HIV-specific antibodies in blood and oral fluid.

To understand test performance and factors contributing to false-negative (FN) results in longitudinal studies, these investigators examined results of participants enrolled in the TDF2 study, the Bangkok Tenofovir Study, and the Bangkok MSM Cohort Study (BMCS) – 3 separate clinical studies of high-risk, HIV-negative persons conducted in Botswana and Thailand. This was a retrospective observational analysis. These researchers compared oral fluid OraQuick (OFOQ) results among participants becoming HIV-infected to results obtained retrospectively using enzyme immunoassay and nucleic acid amplification tests on

stored specimens. They categorized negative OFOQ results as true-negative or false-negative relative to NAAT and/or EIA, and determined the delay in OFOQ conversion relative to the estimated time of infection. These researchers used generalized estimating equations to examine the association between FN results and participant, clinical and testing-site factors. A total of 233 FN OFOQ results occurred in 80 of 287 sero-converting individuals. Estimated OFOQ conversion delay ranged from 14.5 to 547.5 (median of 98.5) days. Delayed OFOQ conversion was associated with clinical site and test operator ($p < 0.05$), pre-exposure prophylaxis ($p = 0.01$), low plasma viral load ($p < 0.02$) and time to kit expiration ($p < 0.01$). Participant age, gender, and HIV subtype were not associated with FN results. Long OFOQ conversion delay time was associated with anti-retroviral exposure and low plasma viral load (PVL). The authors concluded that failure of OFOQ to detect HIV-1 infection was frequent, and multi-factorial in origin. They stated that in longitudinal trials, negative oral fluid results should be confirmed by testing blood samples.

The Aptima HIV-1 Quant Dx Assay

Uprety and colleagues (2020) noted that there are currently no FDA-cleared assays with a dual-claim for both diagnosis and monitoring of HIV-1. The Aptima HIV-1 Quant Dx Assay on the Panther platform (Panther) is the first commercially available test that is CE-marked for both HIV-1 diagnosis and monitoring, but only FDA-cleared for HIV-1 monitoring. These researchers examined the Panther assay for use as a qualitative and quantitative HIV-1 assay in a pediatric population, including patients younger than 24 months old, and reviewed its effect on laboratory efficiency and hands-on-time following its implementation. A total of 100 patient specimens previously tested on the Abbott m2000 RealTime HIV-1 assay (RealTime) and 185 patient specimens previously tested on the Aptima HIV-1 RNA Qualitative Assay (RNA Qual) were tested on the Panther. Verification panels were used to establish precision and linearity. In addition, 268 samples from 134 patients under 24 months of age were also evaluated on the Panther. Overall agreement between the Panther and RealTime assays was 83 %. The mean difference between the 2 methods was 0.10 Log copies/ml. All Panther measurements were linear across the dynamic range ($R^2 = 0.999$). The Panther and RNA Qual assays showed 100 % agreement.

Implementation of the assay opened 600 sq. ft. of space, saved 0.4 full-time equivalent (FTE) and reduced hands-on-time by 70 %. The authors concluded that the Panther assay is an excellent option for HIV-1 qualitative detection and quantitative testing in pediatric patients, including those under 24 months of age. These researchers stated that HIV testing on one platform has opened up space in the clinical laboratory and reduced hands on testing time.

The Reveal G4 Antibody Rapid Test

Rossetti and associates (2020) stated that the Reveal G4 antibody rapid test is FDA-approved for HIV-1 detection using the versions LAB S/P and point-of-care (POC) in CLIA-moderate complexity settings with serum/plasma and whole blood, respectively. The same Reveal tests are CE-marked for HIV-1 and HIV-2 detection in laboratory and POC settings.

These investigators compared the performance of G4 LAB S/P with plasma and POC with whole blood (blood) for detecting early and established HIV-1/HIV-2 infections. Matched well-characterized plasma and simulated blood were used to evaluate: sensitivity in 104 HIV-1 and 55 HIV-2 established infections, specificity in 49 HIV-negative, and reactivity in early HIV-1 infection in a performance panel ($n = 38$) and 18 plasma panels from sero-converters (SCs, $n = 183$). Median number of days after first RNA-positive was calculated for 13 SCs. Impact of viral suppression (VS) was evaluated in 3 SCs receiving early anti-retroviral therapy (ART). Sensitivity was 100 % for HIV-1 and 98.18 % for HIV-2, while specificity was 100 %. All 38 plasma and blood become reactive by Fiebig stage V.

Of 18 SCs, 10 had similar reactivity in plasma/blood, 7 showed delayed reactivity in blood, and 1 was non-reactive in plasma/blood. The median days for a G4-reactive after first RNA-positive was 13 for plasma and 14 for blood. Long-term VS had no impact on G4 reactivity. The authors concluded that overall reactivity in early HIV-1 infections was delayed by 1 day in blood compared to plasma. These researchers stated that if FDA-approved for POC settings, the G4 POC is a fast sensitive screening tool for HIV-1/HIV-2-specific IgG even during VS.

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